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DATA DESCRIPTOR

Dataset of binocularly coded steady-state visual evoked potentials recorded with an augmented reality headset

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Steady-state visually evoked potential (SSVEP)-based brain-computer interfaces (BCIs) have shown significant promise for practical applications. The integration of SSVEP-BCIs with head-mounted augmented-reality (AR) displays is expected to foster wearable, portable systems; nevertheless, empirical resources for such configurations are scarce, especially for paradigms employing innovative stimulation paradigms. Here we present a curated SSVEP dataset recorded with a binocular AR headset that independently modulates the visual input to each eye and a lightweight electroencephalography recorder. Beyond the conventional binocular-congruent single-frequency stimulation adopted in AR-SSVEP studies, the dataset systematically explores binocular-incongruent dual-frequency encoding whereby the two lenses render flickers with distinct frequencies and/or phases. We report comparative analyses of SSVEP characteristics and BCI performance under congruent versus incongruent protocols, and delineate the influence of inter-ocular frequency and phase disparities. The results substantiate the feasibility of wearable AR-SSVEP-BCIs and highlight binocular-incongruent dual-frequency stimulation as a compelling strategy for improving target separability. The dataset should accelerate research on portable SSVEP-BCIs, novel encoding schemes, and the neural mechanisms of binocular vision.

Background & Summary

Brain-computer interfaces (BCIs) allow interaction between the human brain and the external environment without relying on peripheral nerves or muscle tissue and enable users to communicate or manipulate external devices by the signals from the central nervous system^{1–3}. Among various non-invasive BCI paradigms, steady-state visual evoked potential (SSVEP)-based BCIs have been extensively explored due to their non-invasiveness, low hardware cost, high signal-to-noise ratio (SNR), and high information transfer rate (ITR)^{4,5}. SSVEPs are time- and phase-locked electroencephalogram (EEG) signals generated in the occipital region elicited by a frequency-stable visual stimulus^{6–8}. SSVEPs contain strong brain responses at a specific frequency and its harmonics, corresponding to the periodicity of the stimulus⁹. In recent years, SSVEP-BCIs have achieved significant advancements in text spelling^{10,11}, rehabilitation application^{12,13}, device control^{14,15}, etc.

Visual stimulation device is essential for SSVEP-BCIs and is an important factor that can affect the performance, portability and user experience of an SSVEP-BCI system. In existing studies, nearly all of the high-performance SSVEP-BCIs use high-performance liquid crystal display (LCD) or light emitting diode (LED) displays as visual stimulators^{5,10,16}. The SSVEP-BCIs using LCD/LED displays within the frequency range of 6–19 Hz can encode 40–160 targets and achieve ITRs of above 200 bits/min^{17–24}. However, due to the size and weight of LCD/LED monitors, the lack of portability of the SSVEP-BCI using an LCD/LED display severely limits its use in situations where portability is required. Moreover, subjects must constantly shift their gaze between the external device and the visual stimuli on the LCD/LED monitors during the task, which may affect their performance and experience seriously²⁵.

Head-mounted (HM) augmented reality (AR) is a technology that seamlessly integrates virtual scenes into the real environment²⁶ and can provide visual stimulation for SSVEP-BCIs in fully portable and wearable

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devices. The use of HMAR to present visual stimuli enables user interaction with real-world devices by allowing users to view both virtual stimuli and physical objects simultaneously within the same field of view. Additionally, it can dynamically adjust the virtual environment's layout according to the user's current field of view or environmental state²⁷. Consequently, integrating HMAR with an online SSVEP-BCI system allows users to control external devices while interacting with holograms for SSVEP stimulation in real-world settings. For example, Ke *et al.* designed an eight-target AR-SSVEP-BCI that achieved an average ITR of 45.57 ± 7.40 bits/min during an online robotic arm control task²⁸. Hsu *et al.* proposed a six-command AR-SSVEP-BCI to control a drone using a phase-approaching method, achieving an ITR of 36.84 bits/min and a detection accuracy of 93.30%²⁹. Fang *et al.* combined visual information from objects with stimulus targets to adapt to changes in the AR workspace, achieving a detection accuracy of $87.50 \pm 3.10\%$ with a data length of 0.5s³⁰. Zhang *et al.* improved the paradigm, asynchronous pattern recognition, and control logic in an AR-SSVEP-BCI, enabling participants to complete four types of daily life tasks in real-world scenarios³¹. These studies collectively highlight the application potential of AR-BCI, which addresses the limitations of traditional online SSVEP-BCI systems to some extent.

Despite the impressive results achieved by the aforementioned studies, a significant gap remains between the performance of HMAR-based SSVEP-BCIs and that of traditional LCD/LED-based SSVEP-BCIs. Consequently, there is substantial scope for research in AR-based SSVEP-BCIs to explore innovative visual stimulation methods that leverage the intrinsic properties of HMAR. Such exploration could lead to SSVEP-BCI systems with enhanced performance, increased wearability, and improved user experience.

Unlike traditional LCD/LED monitors, A binocular HM display, such as the HMAR, features two displays that present separate images to each eye, thereby generating binocular vision and 3D perception through stereoscopy in the human visual system. This capability of AR devices can be utilized to present different visual stimuli to the left and right eyes separately, such as stimuli with varying frequencies and/or phases for each eye. By using polarization glasses and a circularly polarized light LCD screen, Sun *et al.* developed a binocular dual-frequency SSVEP-BCI system and demonstrated higher quality SSVEP responses and better BCI performance than traditional dual-frequency SSVEP-BCI system by suppressing the unpredictable intermodulation harmonics components (UIHC)^{32,33}. Yang *et al.* reported on a systematical investigation of incongruent dual-, triple-, and quadruple-frequency-coded SSVEP under virtual reality, which confirmed suppression effect of the binocular vision capability and evidenced the potential of binocularly incongruent SSVEP³⁴. Additionally, traditional dual-frequency paradigms that present the same stimulation to both eyes with LCD/LED monitors have demonstrated higher performance and the ability to encode more targets using a limited number of frequencies^{33,35–39}, suggesting the potential of dual-frequency visual stimulation to improve the performance of portable SSVEP-BCIs. Unlike solutions utilizing polarization glasses and circularly polarized light on LCD screens, HMAR devices offer a direct method for implementing binocular disparity visual stimulation and provide a portable, wearable visual stimulation solution for SSVEP-BCIs. However, there remains a paucity of research on binocular-coded SSVEP-BCIs utilizing HMAR stimulators.

To our knowledge, no prior studies have examined the binocular-coded SSVEP-BCIs using HMAR devices. We constructed the first open-access dataset for binocular-coded SSVEP in an HMAR environment using a portable stimulation device. In this study, we proposed an 8-target AR-SSVEP-BCI system and collected EEG data from 30 electrodes located in the parietal and occipital areas. In addition to SSVEPs encoded through single-frequency stimulation of both eyes simultaneously (binocular congruent stimulation) as in previous AR-SSVEP-BCI studies, this study aimed to develop binocular incongruent dual-frequency stimulation methods for multi-target SSVEP-BCIs. This study also compared SSVEP responses and BCI performance between binocular congruent and incongruent stimulations and examines the effects of frequency and phase differences between the two eyes in binocular incongruent stimulation. This dataset aims to provide a comprehensive and high-quality resource for analysing SSVEP features of binocular coding in HMAR environments. It contributes to the development of novel SSVEP encoding and decoding methods, facilitating the study and advancement of wearable and portable SSVEP-BCI systems.

Methods

Participants. The dataset consisted of three experiments: Experiment 1 involved 14 volunteers (10 males and 4 females, aged 20 to 27 years), Experiment 2 also involved 14 volunteers (10 males and 4 females, aged 20 to 27 years), and Experiment 3 involved 17 volunteers (7 males and 10 females, aged 18 to 25 years). All participants had normal or corrected-to-normal vision. None had a history of neurological disease or had taken any medication or received medical treatment in the week preceding the experiment. Each participant was instructed to familiarize themselves with the experimental protocol and sign the consent form, informed of their right to withdraw at any point during the experiment, and agreed to the open publication of experimental data after the study's completion. The experiment was conducted in accordance with the Declaration of Helsinki and was approved by the Research Ethics Committee of Tianjin University (TJUE-2022-047).

Data acquisition. 32-channel EEG data were recorded, with 30 electrodes placed in the parieto-occipital brain region according to the standard positions of the international 10–10 system (PO1, PO2, P1, P5, PO7, CP2, PO8, P6, CP6, CP5, PO5, CP1, PO4, P2, CP7, PO3, POZ, PO6, CP8, CP3, CPZ, CP4, P7, P3, PZ, P4, P8, O1, OZ, and O2) and two electrodes at the left and right mastoids. EEG data were recorded using the Graal 4 K EEG amplifier (Neuroscan, Inc.), a high-performance portable EEG recorder, at a sampling rate of 1024 Hz. The reference electrode was located at Cz, while the ground electrode was placed midway between Fz and FPz. Electrode impedances were maintained below 10 K Ω . As shown in Fig. 1(a), subjects sat in a well-lit room with a wide field of view during the experiment. Figure 1(b) illustrates the data transmission during the EEG recording. Event triggers (stimulus onset) were generated by the stimulation program in the HoloLens 2. Due to its interface limitations, the HoloLens 2 could not be connected directly to the Graal amplifier. We used an embedded

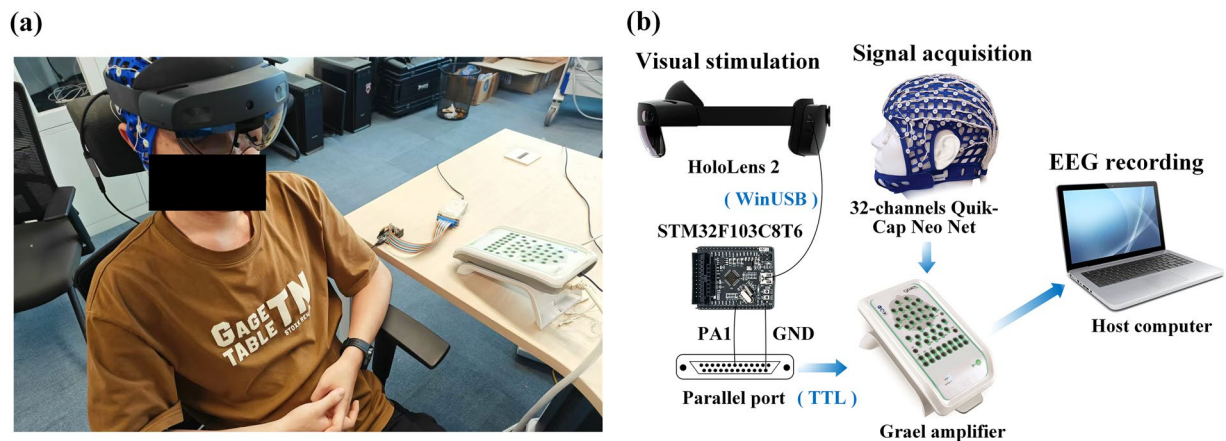


Fig. 1 Schematic diagram of the experimental system. (a) Experimental environment setup. Informed consent was obtained from the participant for publishing the figure. (b) Data transmission during EEG recording.

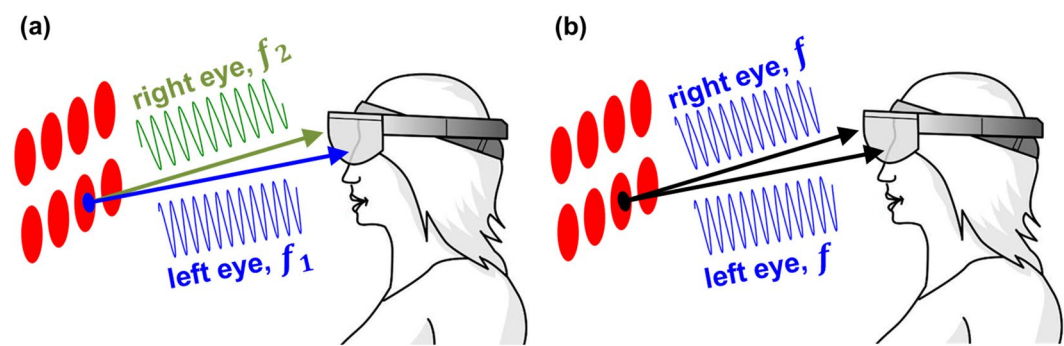


Fig. 2 Schematic illustration of visual stimulus presented to the subject. (a) Binocular incongruent stimulation. (b) Binocular congruent stimulation.

microcontroller (STM32F103C8T6) to transfer the event triggers to the transistor-transistor logic (TTL) port of the Graef amplifier. EEG data and the trigger signals were recorded and saved together on the host computer.

Experimental design. Three experiments were conducted on different days to collect SSVEP data using an HMAR device with binocular congruent coded and binocular incongruent coded visual stimulations. The stimulation interface was developed in C# via Unity 3D (Unity Technologies, San Francisco, CA) and displayed in HoloLens 2 (Microsoft Corporation) with a refresh rate of 60 Hz. By utilizing the HoloLens' multi-pass rendering mode, the content for each lens can be rendered separately, allowing for the display of incongruent visual flickers to each eye through the assignment of distinct layers using the HoloLens SDK within Unity. If a visual stimulus is assigned to the same layer as the eye camera, it can only be seen by the corresponding eye. As shown in Fig. 2(a), a binocular incongruent stimulation can be generated by assigning different layers that contain distinct visual stimuli — coded by varying frequencies and/or phases but sharing identical shape, colour, and position — to the left and right eyes of the subjects at the same position and time. Conversely, Fig. 2(b) illustrates a binocular congruent stimulation, when both eyes are presented with the same stimulation. Subjects wearing the headset perceive a single, unified scene; unless explicitly informed, they are unlikely to notice any inter-ocular differences. Participants were not informed of the differences between experimental conditions beforehand to avoid psychological bias.

Eight targets encoded by the JFPM method⁴⁰ were arranged in a 2×4 matrix set 75 cm away from the subjects in all the experiments (as shown in Fig. 3). The details about the three experiments are as follows.

In Experiment 1, we developed two conditions to assess the feasibility of eliciting SSVEPs through binocular congruent stimulation using low frequency (LF) and medium frequency (MF). The LF ranged from 8 Hz to 15 Hz, with an interval of 1 Hz, whereas the MF ranged from 23 Hz to 30 Hz, also in 1 Hz interval. The phases ranged from 0π to 1.75π with an interval of 0.25π . Regardless of whether LF or MF stimuli, both eyes received congruent stimulation.

In Experiment 2, four conditions were tested to investigate SSVEP responses and BCI performance under various frequency and phase combinations in binocular congruent and incongruent stimulations. In the first condition, eight targets were coded using binocular congruent stimulation with low frequencies, consistent with the LF condition from Experiment 1. Thus, both eyes received visual stimuli at the same frequency and phase (SFSP). In the remaining three conditions, all targets were coded using binocular incongruent stimulation, with

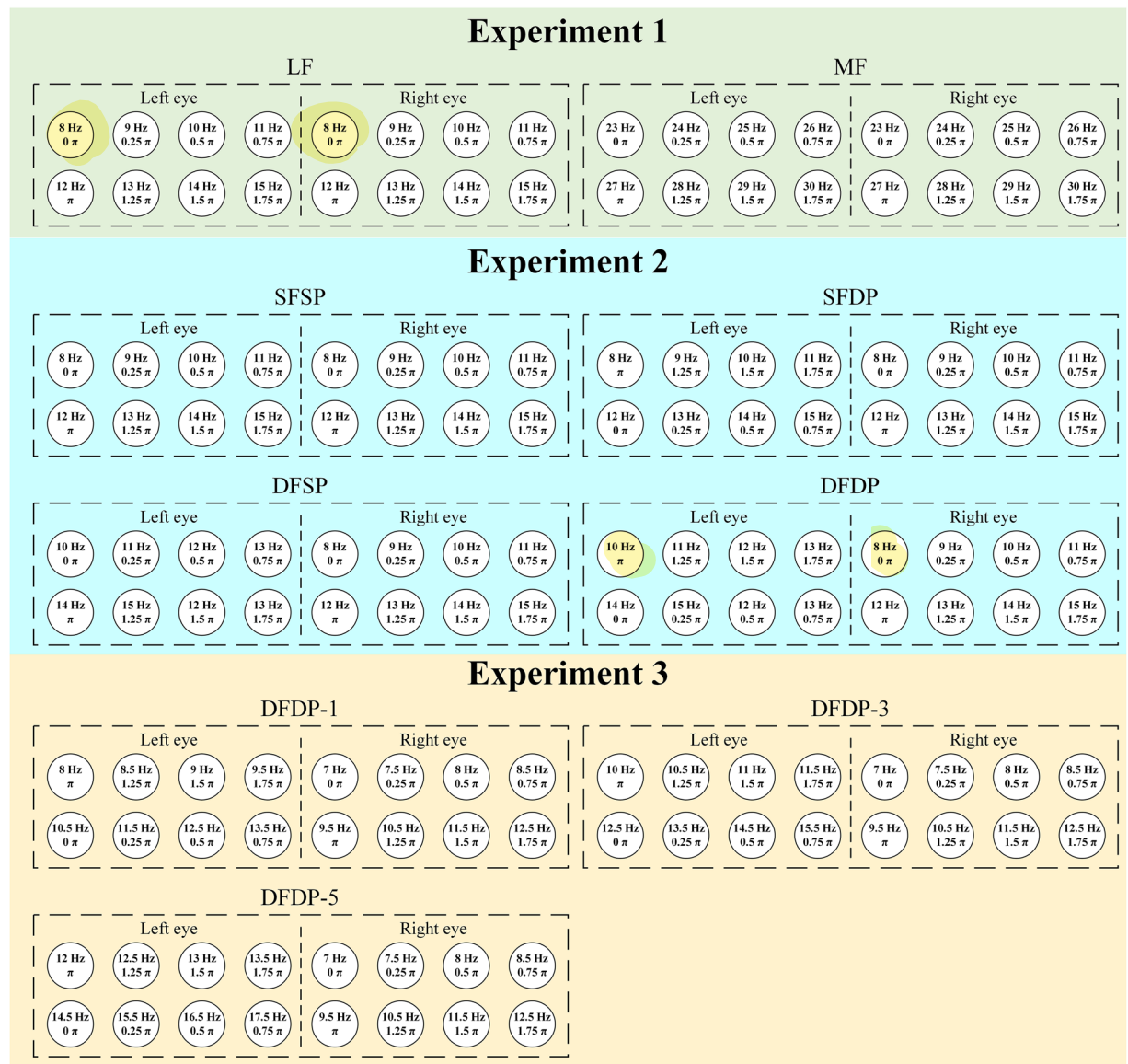


Fig. 3 The stimulus parameters and layout for each condition.

different stimuli presented to each eye. In these three conditions, the stimuli for the two eyes were modulated by two sine waves of the same frequency but different phases (SFDP), different frequencies but the same phase (DFSP), and different frequencies and different phases (DFDP). In the SFDP condition, the stimuli for both eyes were modulated by the same frequency but opposite phases for each target. In the DFSP and DFDP conditions, each target was modulated by two sine waves of different frequencies and either the same or opposite initial phases. **Notably, in both the DFSP and DFDP conditions, two pairs of targets used the same frequency combinations (12 and 14 Hz, 13 and 15 Hz), but the frequencies were swapped between the two eyes. This was done to maintain consistency with the SFSP and SFDP conditions, ensuring that frequency differences did not affect performance comparisons.**

In Experiment 3, three DFDP conditions were conducted to examine the effect of frequency differences between the two eyes, using frequency differences of 1 Hz (DFDP-1), 3 Hz (DFDP-3), and 5 Hz (DFDP-5), with opposite initial phases.

Each condition in the three experiments was conducted in two sub-sessions, with a 5-minute rest period between adjacent sub-sessions. The sub-sessions for all conditions were performed in random order. Each sub-session contained 10 blocks, and each block consisted of 8 trials, comprising 1 trial for each of the eight targets. Every trial consisted of a cue stage and a gaze stage. In the cue stage, a white cross was displayed for 1 second to indicate the upcoming target and the subjects were instructed to turn their gaze to it. In the gaze stage, all eight targets flickered simultaneously for 3 seconds, during which subjects were instructed to gaze at the cued target and to try not to blink.

Data preprocessing. We pre-processed the raw EEG data using the open-source toolbox EEGLAB (<https://scn.ucsd.edu/eeqlab>)⁴¹. A 49–51 Hz notch filter was applied to remove common power line noise, and a band-pass filter was set from 5 Hz to 95 Hz to preserve wideband spectral characteristics. EEG epochs for each trial were extracted according to the event triggers. Due to the latency of the visual system⁴², the time window of the EEG epochs was set to [−0.5, 3.14] seconds, where 0 second indicates the beginning of the flickering.

Performance evaluation. *Signal-to-noise ratio.* The quality of the SSVEP signal could be quantitatively measured by signal-to-noise ratio (SNR)⁴³. Here, a wide-band SNR (measured in decibels, dB) was employed to evaluate the contribution of harmonics. The single-frequency SNR ($\text{SNR}_{\text{single}}$) and dual-frequency SNR (SNR_{dual}) are calculated as follows:

$$\text{SNR}_{\text{single}} = \frac{\sum_{k=1}^{N_h} P(k \cdot f)}{\sum_5^{95} P(f) - \sum_{k=1}^{N_h} P(k \cdot f)} \quad (1)$$

$$\text{SNR}_{\text{dual}} = \frac{\sum_{k=1}^{N_h} [P(k \cdot f_1) + P(k \cdot f_2)]}{\sum_5^{95} P(f) - \sum_{k=1}^{N_h} [P(k \cdot f_1) + P(k \cdot f_2)]} \quad (2)$$

where N_h denotes the number of harmonics, f , f_1 and f_2 denote the frequency of the stimulus target, and $P(f)$ denotes the amplitude at the stimulus frequency f . In this study, N_h was set to 5 for all conditions except MF, where it was set to 3.

Accuracy. The classification accuracy is calculated as the ratio of the number of correctly classified trials to the total number of trials for a given data length. This metric represents the precision of the system and is commonly used to evaluate the performance of BCI systems.

ITR. As a widely used standard in the BCI field, ITR (in bits/min) refers to the number of bits of information transferred per minute. It is calculated as follows:

$$\text{ITR} = \left[\log_2 N + p \log_2 p + (1 - p) \log_2 \left(\frac{1 - p}{N - 1} \right) \right] \cdot \left(\frac{60}{T} \right) \quad (3)$$

where N denotes the number of targets, p denotes the classification accuracy, and T denotes the average target selection time (including the gaze shift time, the visual delay, and the data length used in target identification). In this study, the gaze shift time was set to 1 second.

Data Records

Data format. The experimental records were collected from 24 subjects, de-identified and indexed as S1 to S24. The dataset is openly available on Figshare⁴⁴. To facilitate more comprehensive investigations, the continuous EEG dataset was formatted according to the EEG Brain Imaging Data Structure (BIDS) standard⁴⁵. In accordance with the standard, the data of each participant are organized within a designated folder (e.g., “sub-001”). The raw EEG data are stored in “.set” and “.fdt” files, while the “.tsv” and “.json” files contain relevant information about the experiment. A preview of the data structure is provided in Fig. 4.

Experiment information. The participation of all subjects in the experiment is listed in Table 1. Additional information is stored in the “participants.tsv” file, which includes non-identifiable participant information such as age and gender. Information on the electrodes can be found in the “*_electrodes.tsv” file, which includes electrode details such as name, index, position in polar coordinate, and impedance value.

Technical Validation

Since the SSVEP response is concentrated in the parietal and occipital areas, ten electrodes (PO7, PO5, PO3, POz, PO4, PO6, PO8, O1, Oz, and O2) are used in the following analysis.

SSVEP signal characteristic analysis. We validated our dataset by examining time- and frequency-domain characteristics. The Oz channel was chosen as the representative channel for visual inspections. To enhance the quality of the SSVEP signal, we performed a grand average of EEG data across blocks and subjects in each condition. Figure 5 presents the waveforms of a target in the time domain for all experimental conditions. Figure 6 displays the corresponding frequency-domain responses derived from these waveforms. Giani *et al.* demonstrated that the human brain generates within-modality integration in response to various sensory stimuli⁴⁶. Both eyes received the stimulus at the same frequency but in opposite phases, leading to negative within-modality integration, which reduced the amplitude of the response. Additionally, Giani *et al.* revealed that the response to dual-frequency stimulation in the time domain is a modulated superposition produced by a non-linear system⁴⁶. In the SFSP conditions (LF and MF in Experiment 1, and SFSP in Experiment 2), two simultaneous visual flickers share the same fundamental frequency and are presented in phase (0° phase offset). Because the phase-locked cortical responses are perfectly aligned, their linear superposition produces constructive interference. The resulting SSVEP therefore exhibits a strong, temporally stable oscillation with an enhanced fundamental amplitude and negligible fluctuation, reflecting the coherent summation of the two synchronized neural responses. In the SFDP condition, two visual flickers of identical fundamental frequency are presented in



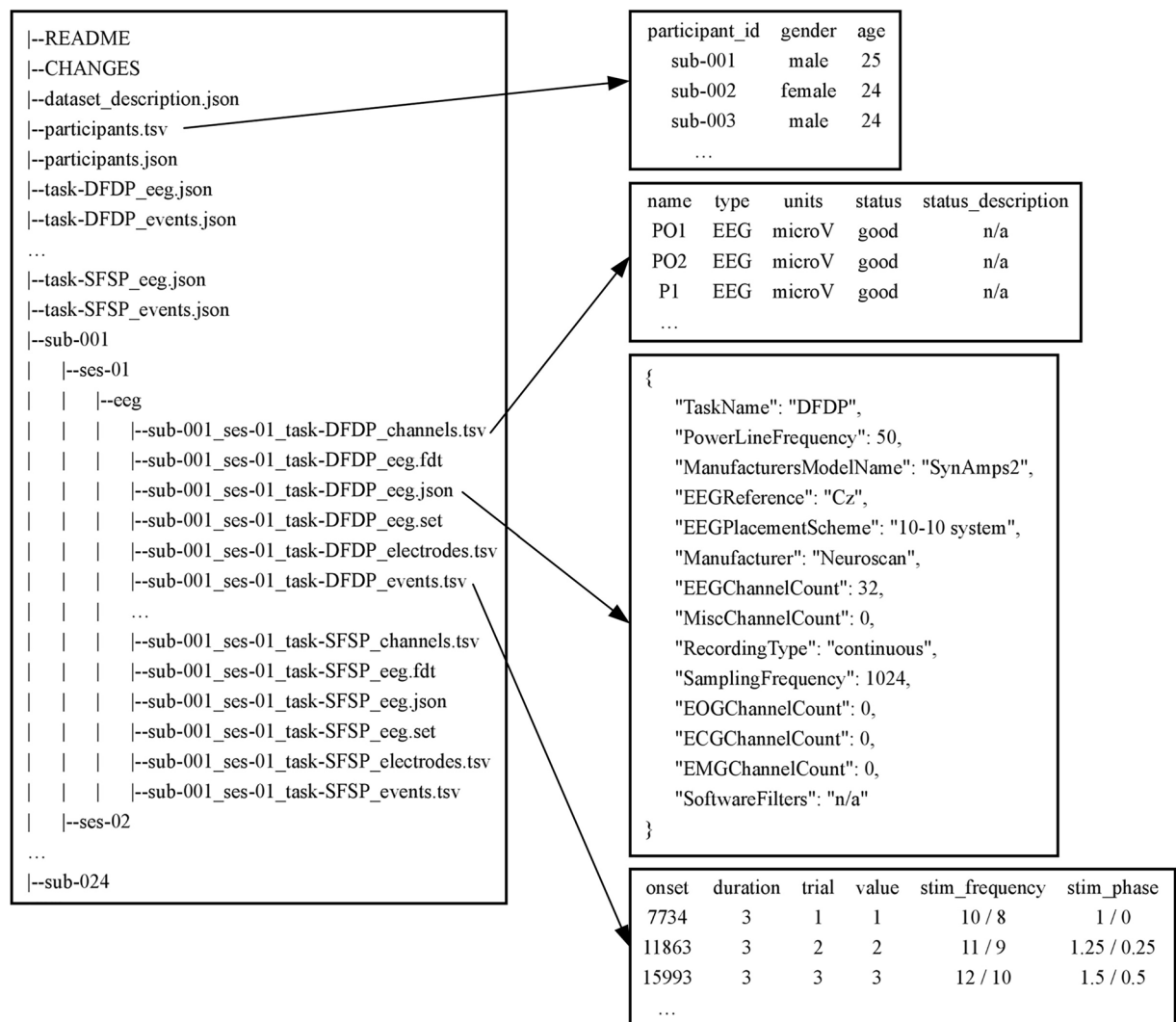


Fig. 4 A preview of raw EEG data records of the database. The data records were curated according to the EEG-BIDS⁴⁵. The raw EEG were stored in the EEGLAB format (".set" and ".fdt"). The prefix "sub" denotes the participant and "ses" denotes the session.

antiphase (180° phase offset). Because the evoked cortical responses at that frequency are phase-locked yet opposite in phase, their linear superposition leads to destructive interference. Consequently, the SSVEP exhibits a marked attenuation of the fundamental component: the amplitude is substantially reduced as the two antiphase oscillations cancel each other, yielding a diminished SSVEP magnitude despite ongoing stimulation at the same frequency. In the DFSP and DFDP conditions, two concurrent visual flickers at distinct fundamental frequencies (f_1 and f_2) entrain the visual cortex simultaneously. The resulting SSVEP is therefore not a single sinusoid but the linear superposition of the two driven oscillations together with their nonlinear interaction terms. This superposition produces a pronounced temporal-interference (beat) phenomenon: the carrier components at f_1 and f_2 remain clearly visible in the spectrum (see Fig. 6), while their vector summation in the time domain yields an amplitude envelope that oscillates at the beat frequency $|f_1 - f_2|$. The envelope modulation reflects periodic constructive and destructive interference between the two phase-locked neural responses, generating cyclic enhancements and suppressions of the SSVEP amplitude over time. Accordingly, DFDP-1 produces a slow envelope (1 Hz), DFDP-3 a faster envelope (3 Hz), and DFDP-5 the fastest envelope (5 Hz), confirming that the beat frequency scales directly with the separation of the driving frequencies.

Figure 6 displays the 6–70 Hz amplitude spectra derived from the time-domain waveforms in Fig. 5. Under monocularly matched, single-frequency stimulation (LF, MF, SFSP), the spectra show the canonical SSVEP profile: a dominant fundamental peak accompanied by progressively weaker harmonics, reflecting constructive binocular summation. When the same frequency is delivered in antiphase (SFDP), both the fundamental and its harmonics are greatly attenuated, indicating destructive interference between the phase-opposed cortical responses. Dual-frequency, binocularly incongruent stimulation (DFSP, DFDP) elicits two distinct fundamental peaks and their harmonic series, together with multiple intermodulation components. Relative to the SFSP condition, the fundamental amplitudes are reduced, whereas higher-order sum- and difference-frequency

Subject	Experiment 1	Experiment 2	Experiment 3
S1	✓	✓	✓
S2	✓	✓	✓
S3	✓	✓	✓
S4	✓	✓	✓
S5	✓	✓	✓
S6	✓	✓	✓
S7	✓	✓	✓
S8	✓	✓	✓
S9	✓	✓	×
S10	✓	✓	×
S11	✓	✓	×
S12	✓	✓	×
S13	✓	✓	×
S14	✓	×	×
S15	×	✓	×
S16	×	×	✓
S17	×	×	✓
S18	×	×	✓
S19	×	×	✓
S20	×	×	✓
S21	×	×	✓
S22	×	×	✓
S23	×	×	✓
S24	×	×	✓

Table 1. The participation of all subjects.

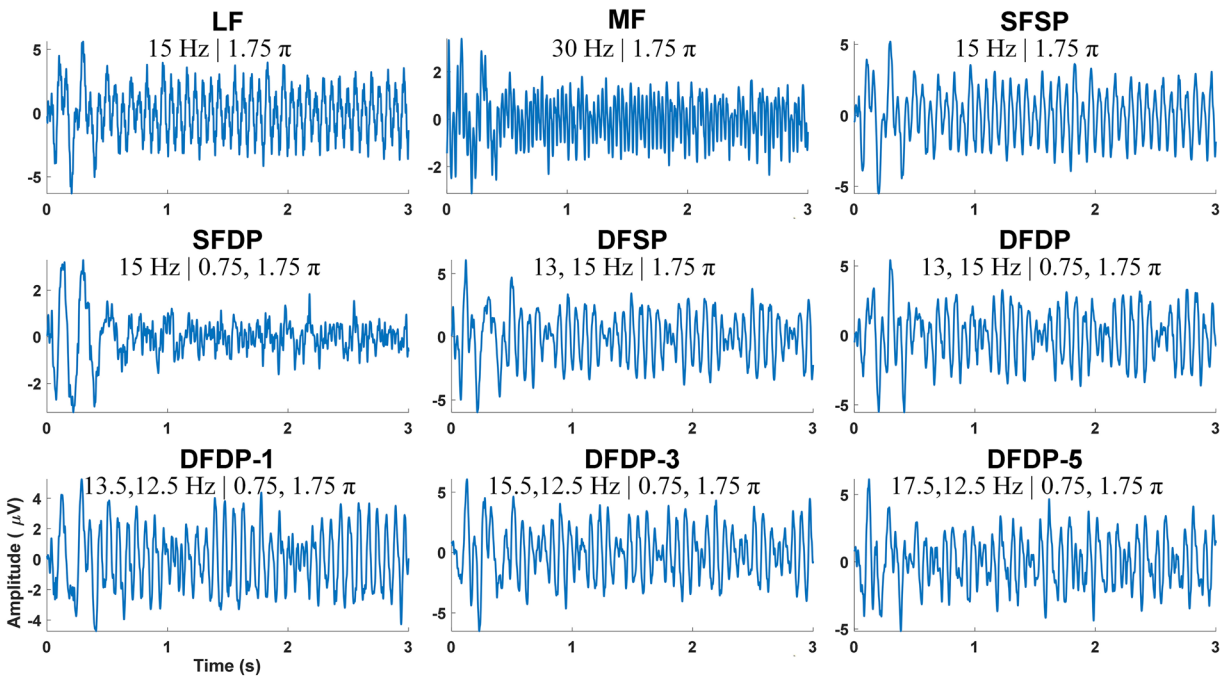


Fig. 5 The time-domain waveforms of a target for all conditions.

components ($a * f_1 + b * f_2$) become more prominent, signifying enhanced nonlinear binocular interactions in the dual-frequency paradigm. In the graded antiphase conditions (DFDP-1, -3, -5; $\Delta f = 1, 3, 5$ Hz), the spacing between the two fundamental families scales precisely with the imposed frequency difference, and the associated intermodulation clusters disperse as Δf increases. Collectively, these spectra indicate that interocular phase determines whether constructive or destructive interference occurs at the driving frequencies, while the

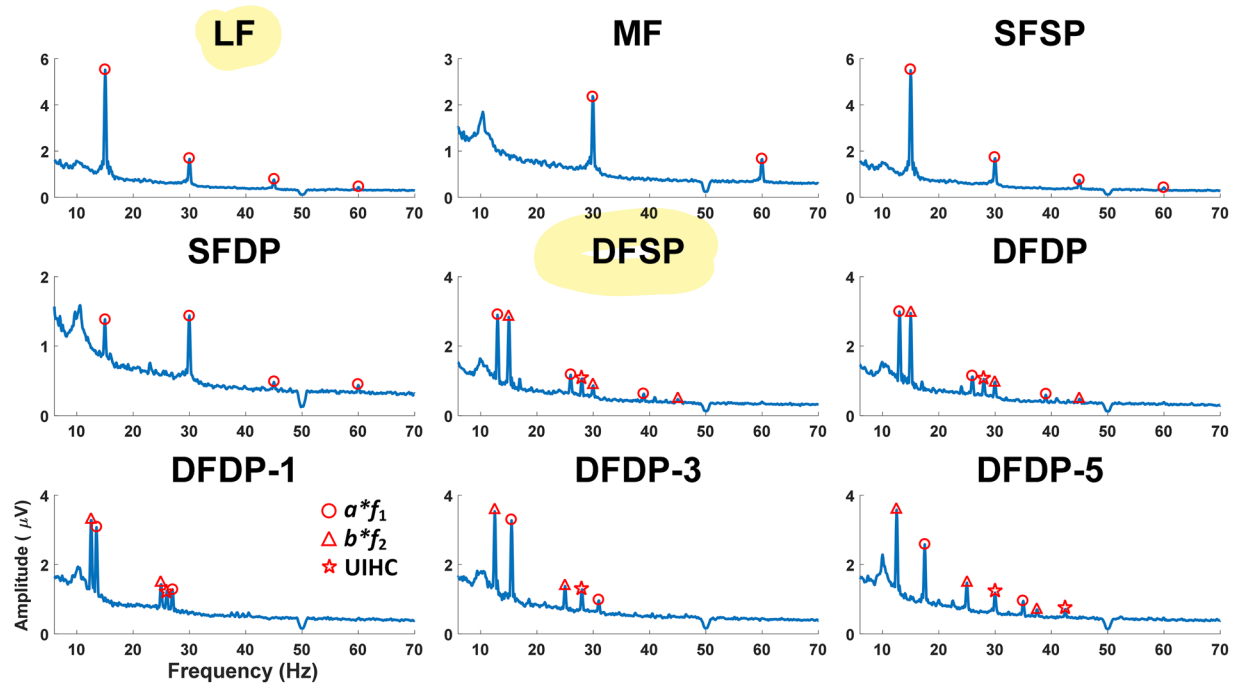


Fig. 6 The frequency responses of the target with the highest stimulus frequency.

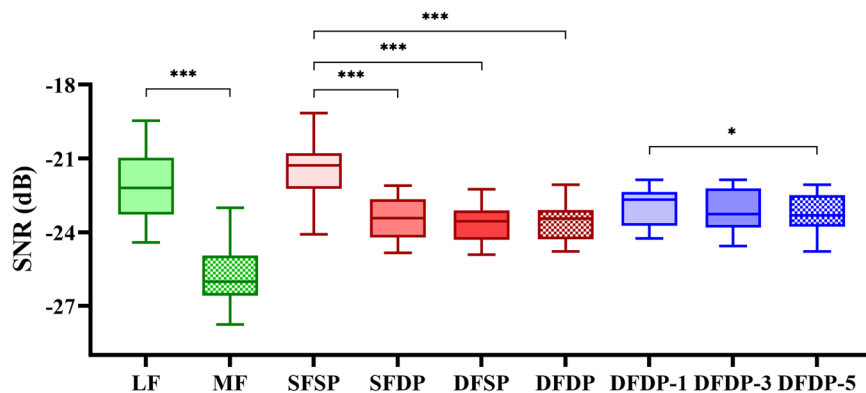


Fig. 7 The wide-band SNRs for all conditions. Asterisks indicate significant difference after correction for multiple comparisons (***) $p < 0.001$, (*) $p < 0.05$.

combination of phase and frequency separation dictates the magnitude and distribution of nonlinear interaction components in binocular SSVEPs.

Previous studies define UIHC, generated by dual-frequency stimulation, as spectral lines at $a * f_1 + b * f_2$, where a and b are arbitrary integers^{32,33}. Owing to their instability and inter-individual variability, UIHC can degrade classification accuracy in conventional LCD/LED-based SSVEP BCIs. In the present study, however, each eye receives an independent stimulus in dual-frequency conditions, preventing the frequencies from overlapping before they reach the optic nerve. This separation markedly suppresses UIHC, making such components far less prevalent. Nonetheless, the sum frequency $f_1 + f_2$ persists in all dual-frequency conditions, in agreement with the observations of Sun *et al.*^{33,39}.

To characterise the SNR properties of the dataset, we computed the SNR for each experimental condition. The amplitude spectra of the three-second epochs from the ten occipital-parietal channels were first averaged, after which the SNR was quantified for each condition by averaging across targets. The resulting SNRs for all conditions are illustrated in Fig. 7. Paired t-test was conducted for each experiment, and Bonferroni corrections were applied to account for multiple comparisons. Significant differences are indicated by asterisks in Fig. 7. For Experiment 1, the LF paradigm produced significantly higher SNR than the MF paradigm ($p < 0.001$). For Experiment 2, the SNR was significantly higher (all $p < 0.05$) for the binocular-congruent paradigm (SFSP) compared to the binocular-incongruent paradigms (SFDP, DFSP, DFDP). For Experiment 3, a significant difference was observed only between DFDP-1 and DFDP-5 ($p < 0.05$). In addition, the SNR distributions lie within

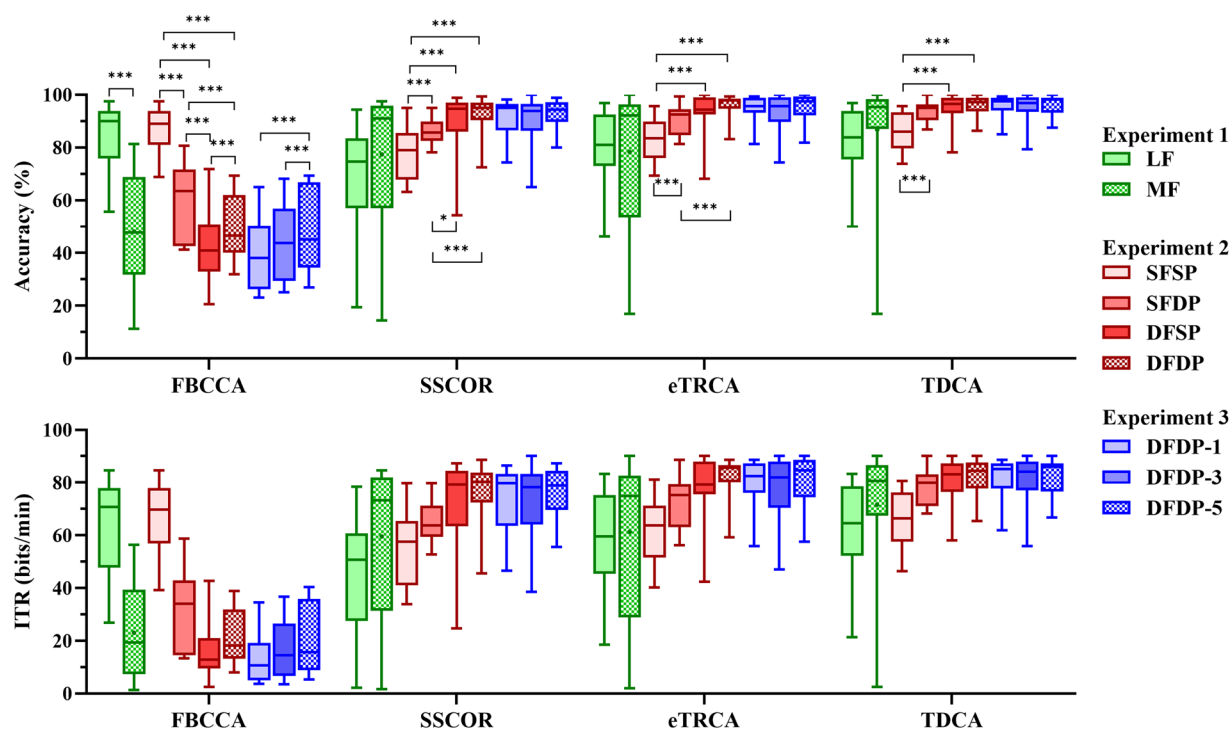


Fig. 8 The classification accuracy for each condition calculated by different algorithms with a data length of 1 second. Asterisks indicate significant difference after correction for multiple comparisons (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$).

the expected range reported in the literature⁴⁷, indicating that the recordings contain a robust evoked component relative to the background EEG.

Taken together, the time-domain waveforms, frequency responses, spectral profiles, and SNR measures confirm the overall reliability and quality of the present EEG dataset.

Classification analysis. To validate the utility of this dataset, classification analysis was conducted by four recognition algorithms: filter bank canonical correlation analysis (FBCCA)⁴⁸, sum of squared correlations (SSCOR)⁴⁹, ensemble task-related component analysis (eTRCA)⁵⁰, and task-discriminant component analysis (TDCA)⁵¹. We employed leave-one-block-out cross-validation with different data lengths ranging from 0.1 to 3 seconds, with an interval of 0.1 second. The filter-bank method was applied in each algorithm with the number of filter banks set to $N_{fb} = 5$ (for all conditions except MF) or $N_{fb} = 3$ (for MF) and the band weights were set according to a previous study⁴⁸. The classification accuracy of all subjects for each condition, calculated using different algorithms with a data length of 1 second, are shown in Fig. 8.

Two-way repeated-measures ANOVAs with the factors Paradigm and Algorithm were conducted separately for each experiment; Geisser–Greenhouse corrections were applied when the assumption of sphericity was violated. A significant Paradigm $\times$ Algorithm interaction was observed in all three experiments (Experiment 1: $F(1.97, 25.65) = 23.1$, $p < 0.0001$; Experiment 2: $F(3.40, 44.21) = 120.8$, $p < 0.0001$; Experiment 3: $F(1.21, 19.32) = 8.5$, $p = 0.0065$). Consequently, post-hoc pairwise comparisons were performed for the factor Paradigm within each Algorithm, and significant differences are marked by asterisks in Fig. 8. For Experiment 1, Šidák-adjusted tests showed that the LF paradigm produced significantly higher classification accuracy than the MF paradigm when the FBCCA algorithm was used ($p < 0.05$). No significant differences emerged with the remaining algorithms. For Experiment 2, Tukey-corrected tests revealed a clear ordering of accuracies under FBCCA (SFSP > SFDP > DFDP > DFSP; all $p < 0.05$). With the other three algorithms, the ordering was SFSP > SFDP > DFSP > DFDP. In these algorithms, binocular-incongruent paradigms (SFDP, DFSP, DFDP) outperformed the binocular-congruent paradigm (SFSP), and dual-frequency stimulation (DFSP, DFDP) yielded higher accuracies than single-frequency stimulation (SFSP, SFDP). For Experiment 3, significant paradigm effects were detected only for FBCCA (Tukey-corrected, $p < 0.05$); no pairwise differences reached significance with the other algorithms. These results demonstrate that SSVEP decoding accuracy is shaped by a pronounced interaction between stimulation paradigm and decoding algorithm. For the supervised methods examined (SSCOR, eTRCA, and TDCA), binocular-incongruent paradigms consistently outperformed binocular-congruent paradigms, and this advantage was unaffected by the frequency separation between the two component flickers. These findings conflict with the SNR analysis, which favours the congruent stimuli. One possible explanation is that time-domain waveform characteristics—such as inter-ocular interference—exert a greater influence on template-matching-based SSVEP recognition than the frequency-domain power captured by SNR. Additional research is needed to elucidate this discrepancy.

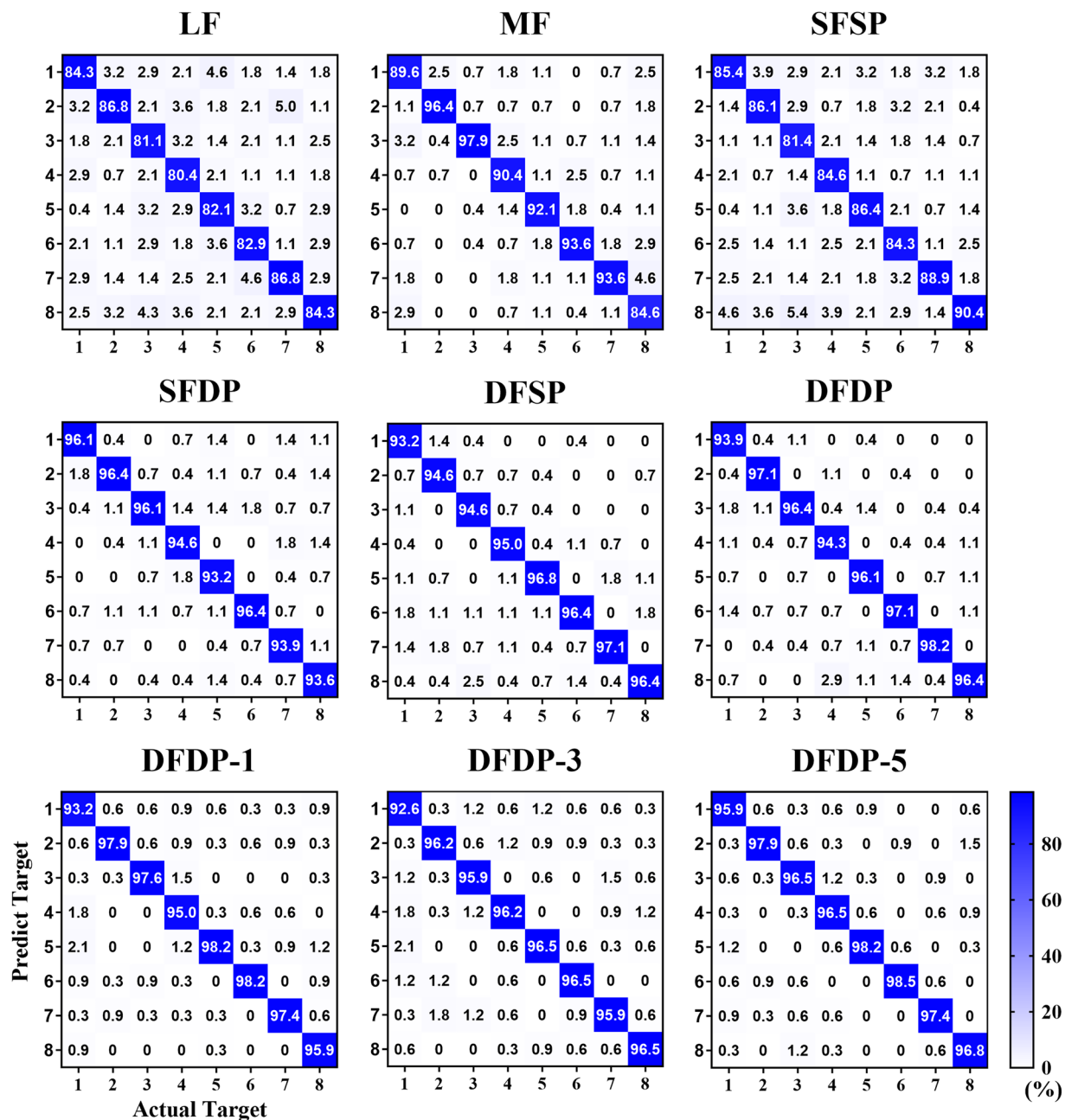


Fig. 9 Averaged 8-class confusion matrices calculated by TDCA with a data length of 1 second.

Because TDCA yielded the highest accuracies across paradigms, we computed mean confusion matrices with this algorithm. Figure 9 presents the resulting 8-class matrices obtained from 1-s epochs. In the binocular-congruent conditions, all targets were classified with accuracies above 81%; in the binocular-incongruent conditions, accuracies exceeded 92%. Importantly, in Experiment 2 the DFSP and DFDP paradigms contained target pairs (5 vs 7 and 6 vs 8) that shared identical frequency combinations, yet the classifier still distinguished them reliably. These findings illustrate that binocular-incongruent stimulation can encode a larger number of commands with a limited frequency set while preserving high decoding performance.

Figures 10, 11 summarise how decoding accuracy and information-transfer rate (ITR) vary with the length of the analysis window (0.1–3.0 s, in 0.1-s increments). Across all stimulation paradigms and algorithms, accuracy rises sharply during the first 0.5 s and then approaches an asymptote, with only minor gains beyond 1 s. Conversely, ITR reaches its maximum between 0.4 and 0.6 s and subsequently declines. The binocular-incongruent conditions—DFSP, DFDP, and the three graded variants DFDP-1/-3/-5—consistently yield the most favourable trade-off between speed and reliability: accuracies exceed 92% after only 0.5 seconds of data, and the accompanying peak ITR reaches 90–105 bits/min (TDCA), marginally surpassing the binocular-congruent condition (SFSP). In every paradigm in Experiments 2 and 3, the supervised methods (TDCA > eTRCA > SSCOR) markedly outperform FBCCA.

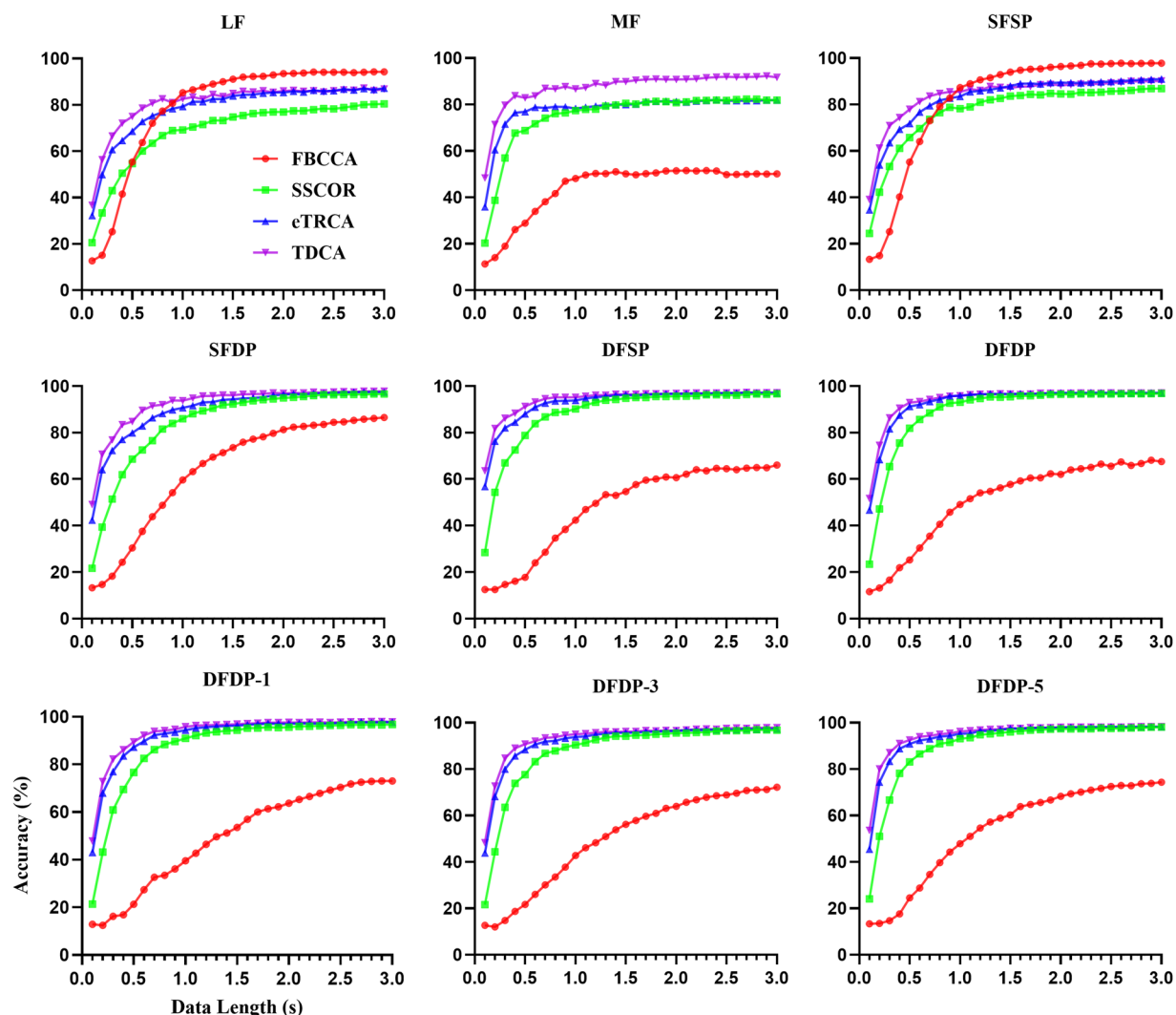


Fig. 10 Average classification accuracy was evaluated across analysis windows ranging from 0.1 s to 3.0 seconds, using 0.1-second increments.

This study introduces a novel binocular-incongruent stimulation paradigm for HMAR. As outlined in the Introduction, HMAR enables users to perceive virtual stimuli and real-world objects within a single field of view, thereby supporting more natural and continuous interaction. To the best of our knowledge, this work represents the first systematic examination of binocular-incongruent, multi-frequency SSVEP coding in an HMAR setting, including implementation procedures and an analysis of the resulting EEG signatures. Data were acquired with portable EEG hardware and subjected to comprehensive feature and classification analyses, confirming the high quality and stability of the recordings. The results demonstrate that binocular-incongruent stimulation can encode a larger number of targets with a limited frequency set, and that left–right frequency swapping further augments this benefit. To facilitate future research on the neural mechanisms underlying this phenomenon and to support the development of improved decoders, we have made the full binocular-coded SSVEP dataset publicly available. Unlike existing public dual-frequency SSVEP datasets—which were collected on static LCD/LED screens presenting identical stimuli to both eyes^{35–39} or using bulky polarization setups^{32–34}—our HMAR solution delivers truly independent left- and right-eye flickers within a fully wearable AR + EEG platform. This design not only captures SSVEPs in mobile conditions but also spans congruent and systematically varied incongruent frequency- and phase-offset protocols. Initial analyses show that binocular-incongruent stimulation in this dataset yields superior performance in target classification and inter-ocular temporal-interference features, providing a richer benchmark for portable SSVEP-BCI research.

Usage Notes

We provide raw data in this study for offline analysis or online simulation. For analysing the dataset, we recommend using common open-source toolboxes for EEG data, such as EEGLAB⁴¹ in the MATLAB environment, or MNE⁵² in the Python environment. Leave-one-block-out cross-validation is a common method in classification analysis. Due to the temporal and sequential correlations of stimuli, data partitioning by trials or using random sliding windows should be avoided. This dataset can be used to analyse SSVEP features of binocular-coding

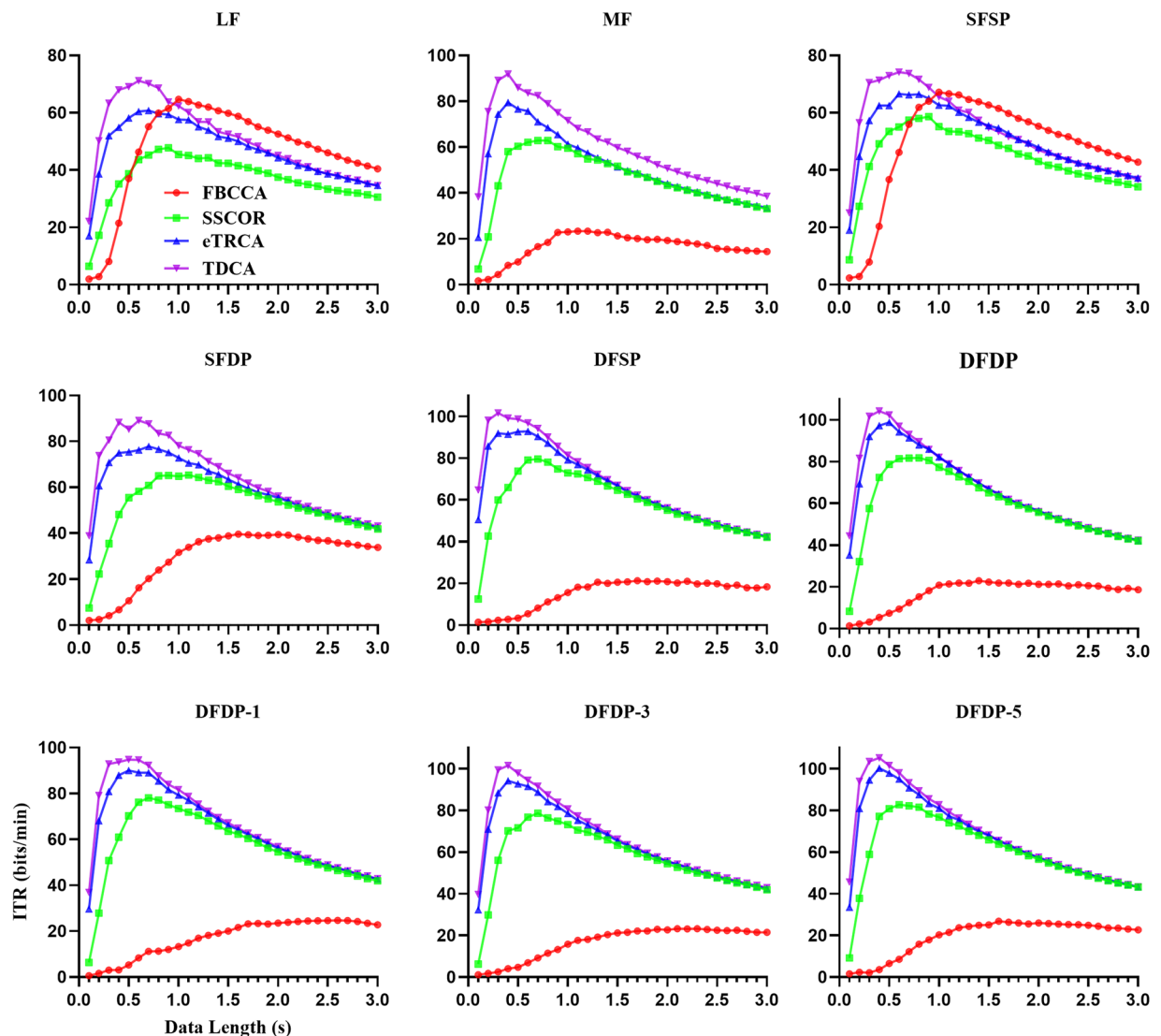


Fig. 11 Average ITR was evaluated across analysis windows ranging from 0.1 s to 3.0 seconds, using 0.1-second increments.

stimulation with quantitative evaluation and to evaluate the performance of different SSVEP-BCI classifiers in the AR environment.

Code availability

The custom MATLAB codes for loading data, evaluating classification performance and signal quality, and plotting figures can be found in the repository⁵³. The function of “LoadData.m” is to pre-process the data and save it as 4-dimensional matrices (target $\times$ channel $\times$ sampling point $\times$ block). Codes under the “Classification” folder are presented to validate data quality and algorithm performance, which utilize publicly available code with references noted^{48–51}. At the top of the code, there are options to set the parameters of the analysis, such as setting the path with the variable “Path”, selecting participants, conditions and algorithms with the variables “subject”, “stimuli” and “algorithm”, and choosing data length with the variable “Timelength”. The function of “PlotFigure.m” is to analyse and graph EEG response features. A brief description of the code is summarized in “README.txt” to assist researchers in running the code.

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Author contributions

Y.K. contributed to the design of the work, data collection, data validation and manuscript preparation. Y.H. contributed to data validation and manuscript preparation. P.L. contributed to the design of the experiment, data collection. D.M. supervised this project and contributed to the conception and design of the work.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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